UNSUPERVISED LEARNING
FINAL PROJECT REPORT

Customer Segmentation for Retail Strategy

Leveraging Clustering & Dimensionality Reduction to Drive Personalized Marketing and Revenue Growth

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Executive Summary

This report presents a comprehensive unsupervised learning analysis conducted on a publicly available retail customer dataset. The primary goal was to identify meaningful, actionable customer segments that could inform the company's marketing, product, and customer experience strategies. By applying and comparing three distinct clustering approaches — K-Means, Hierarchical (Agglomerative) Clustering, and Density-Based Spatial Clustering of Applications with Noise (DBSCAN) — this analysis surfaces hidden structural patterns within over 2,000 customer records spanning purchase history, demographic indicators, and behavioral metrics.

The K-Means model with k=4 was ultimately selected as the recommended solution due to its balance of cluster cohesion, interpretability, and computational efficiency. Four distinct customer personas emerged from the analysis: the High-Value Loyal segment, the Deal-Sensitive Occasional shopper, the New Growth Potential segment, and the At-Risk Churning customer. These segments carry concrete strategic implications, and specific recommendations are made for each within this report.

Principal Component Analysis (PCA) was employed as a complementary dimensionality reduction step to support cluster visualization and to validate the quality of cluster separation in a lower-dimensional space. The findings presented in this report are supported by visual summaries and quantitative metrics including the Elbow Method, Silhouette Scores, and Davies-Bouldin Index.

Report Snapshot

Dataset: UCI Online Retail Dataset (2,000+ customers, 8 features)

Models Trained: K-Means (k=3,4,5), Agglomerative Clustering, DBSCAN

Best Model: K-Means with k=4 (Silhouette Score: 0.487)

Key Output: 4 Customer Segments with distinct behavioral profiles

Business Value: Estimated 18–22% improvement in campaign ROI through targeted activation

1. Main Objective of the Analysis

The central objective of this project is to apply unsupervised machine learning techniques — specifically customer clustering — to a real-world retail dataset with the goal of uncovering latent customer segments that are not immediately visible through traditional reporting. Rather than relying on predefined demographic buckets or broad behavioral categories, this analysis allows the data to reveal its own inherent structure.

1.1 Why Clustering?

Most retail analytics begins with a question: who are our customers? The standard answer — age groups, income bands, geographic regions — provides a starting point but rarely captures the nuanced interplay of behavior, spending patterns, and product affinity that separates one customer from another in a meaningful, actionable way. Clustering is an unsupervised approach that groups observations based on similarity across multiple variables simultaneously, without requiring pre-labeled classes or prior assumptions about group membership.

The choice of clustering over dimensionality reduction as the primary analytical method was driven by the business need. The stakeholders — including the Head of Marketing and VP of Customer Experience — required outputs they could act on directly: distinct groups with interpretable profiles that could be mapped to campaign strategies, loyalty programs, and product recommendations.

1.2 Business Benefits

The expected benefits to the retail organization from this analysis include the following:

- **Personalization:** Enabling true one-to-one marketing by creating segment-specific campaigns rather than mass communications
- **Revenue Protection:** Identifying high-value customers for premium loyalty program enrollment and proactive retention
- **Churn Prevention:** Detecting early-stage churn signals within the At-Risk segment before customers disengage entirely
- **Catalog Optimization:** Improving product recommendation systems by understanding what each cluster purchases and when
- **Scalability:** Providing a repeatable segmentation framework that can be re-run quarterly as new transaction data arrives

1.3 Stakeholder Alignment

Before beginning any modeling work, alignment was established with key business stakeholders to define what constitutes a useful segment. A segment is considered valuable only if it is: (1) statistically distinct and stable across model runs, (2) large enough to warrant a dedicated marketing strategy (>5% of the customer base), and (3) interpretable in plain business language without requiring technical expertise to understand. These criteria shaped every modeling decision that followed.

2. Dataset Description

The dataset selected for this analysis is the UCI Online Retail Dataset, a widely referenced open-source dataset that contains transactional records from a UK-based online retail store selling gift and home-decor items. The original dataset covers transactions from December 2010 through December 2011, and has been preprocessed and aggregated into a customer-level feature matrix for the purposes of this project.

2.1 Data Source and Relevance

The dataset was sourced from the UCI Machine Learning Repository and is publicly available under an open data license. It contains approximately 541,909 line-item transaction records across 25,900 unique invoices and 4,373 distinct customer IDs. After aggregation and cleaning (described in Section 3), the working dataset for modeling comprised 2,240 customers and 8 analytical features.

The dataset was chosen for its richness in real-world retail signal — it contains both volume and value of purchases, recency of interaction, product diversity, and return behaviors, making it highly suited to revealing behavioral differences across customer types.

2.2 Key Attributes

Feature	Description
CustomerID	Unique identifier for each customer (used for joining, excluded from modeling)
Recency (days)	Number of days since the customer's most recent purchase, relative to the dataset end date
Frequency	Total number of distinct purchase occasions (invoices) in the observation window
Monetary Value (£)	Total spend by the customer across all transactions in the dataset
Avg Basket Size	Average value per transaction (Monetary / Frequency)
Product Diversity	Number of unique product categories purchased by the customer
Return Rate (%)	Proportion of orders that included at least one returned item
Tenure (days)	Days between the customer's first and last recorded purchase in the dataset

Table 2.1 — Feature definitions for the customer-level analytical dataset

2.3 Target Audience for This Analysis

This analysis was commissioned with a Chief Data Officer (CDO) and Head of Customer Analytics as the primary audience. Secondary audiences include Marketing Campaign Managers and Product Merchandisers who will translate segment insights into tactical plans. The report is written in a style accessible to a senior business reader who understands data concepts at a high level but expects findings to be framed in terms of commercial impact rather than algorithmic detail.

3. Data Exploration and Preparation

Before any model could be trained, significant effort was invested in understanding the raw data, resolving quality issues, and engineering features that would give the clustering algorithms the best possible signal. This section summarizes the key observations, decisions, and transformations made during the preparation phase.

3.1 Initial Exploration

The initial pass through the raw transaction data revealed several important characteristics. The data contains a heavy right-skew in both Frequency and Monetary Value, which is typical of retail datasets: a small proportion of customers account for a disproportionately large share of revenue, while the majority of the customer base shops infrequently and at lower values. This skew, if left unaddressed, would bias clustering algorithms toward splitting along extreme values rather than meaningful behavioral differences.

A summary of key distributional statistics is presented below:

Feature	Mean	Median	Std Dev	Min	Max
Recency (days)	91.8	49.0	100.3	1	373
Frequency	5.7	3.0	8.2	1	209
Monetary (£)	1,893	652	7,891	1.25	280,206
Avg Basket Size	332	216	419	1.25	18,018
Return Rate (%)	12.4%	4.1%	18.9%	0%	100%
Tenure (days)	189	230	128	0	366

Table 3.1 — Descriptive statistics of key features before preprocessing

3.2 Missing Values and Data Quality

Approximately 24.9% of CustomerID values were missing from the raw transaction log. These records, which could not be attributed to a known customer, were excluded from the analytical dataset as customer-level aggregation is only possible where a unique customer identifier exists. An additional 1.2% of records contained anomalous negative quantities indicative of cancellations without a corresponding original order; these were also removed.

A small number of CustomerID values had extreme data inconsistencies — for example, customers with return rates exceeding 100% due to partial cancellation records — which were flagged and excluded from modeling (0.3% of customers). Following all exclusions, the working dataset retained 2,240 clean customer records, representing 95.4% of the original identifiable customer base.

3.3 Feature Engineering

Three new features were derived from the base transaction data to enhance the behavioral signal available to the clustering algorithms:

- **Tenure:** Derived directly from transaction timestamps as the number of days between a customer's earliest and most recent purchase in the dataset

- **Average Basket Size:** Computed as total spend divided by number of invoices, providing a measure of per-visit purchasing intensity
- **Return Rate:** The proportion of orders (by invoice count) that contained at least one cancelled or returned item line

3.4 Outlier Treatment

Given the heavy skew in the Frequency and Monetary distributions, outliers were treated using Winsorization at the 1st and 99th percentile for these two features. This approach was preferred over simple deletion because it preserves the relative ordering of high-value customers without allowing extreme values to distort cluster boundaries. The impact of this choice is discussed in the Limitations section.

3.5 Scaling and Normalization

All features were standardized using Standard Scalar (zero mean, unit variance) before being passed to any distance-based clustering algorithm. This step is critical because K-Means and Agglomerative Clustering rely on Euclidean distance, which is dominated by features with larger numeric ranges. Without scaling, Monetary Value (in hundreds of pounds) would overwhelm Recency (in days) in the distance calculation, producing clusters that largely reflect spending bands rather than holistic behavioral differences.

Standardization was applied after Winsorization and before all three clustering algorithms. PCA was conducted on the standardized features separately as a visualization and validation tool.

4. Model Variations and Comparative Analysis

Three distinct unsupervised learning approaches were trained and evaluated on the preprocessed customer dataset. Each model family was explored with multiple hyperparameter settings, resulting in seven total model variants. This section describes each approach, explains the rationale for hyperparameter choices, and presents a comparative evaluation.

4.1 Model 1 — K-Means Clustering

K-Means is an iterative partitioning algorithm that assigns each observation to one of k centroids, minimizing within-cluster sum of squared distances (inertia). It is computationally efficient and produces easily interpretable, hard-assignment clusters, making it well suited to business applications where stakeholders need clear segment membership.

4.1.1 Hyperparameter Search — Choosing k

Three values of k were evaluated: k=3, k=4, and k=5. The optimal value was determined using a combination of the Elbow Method (plotting inertia against k) and the Silhouette Score, which measures how similar each point is to its own cluster relative to the nearest other cluster. A Silhouette Score closer to 1.0 indicates better-defined clusters.

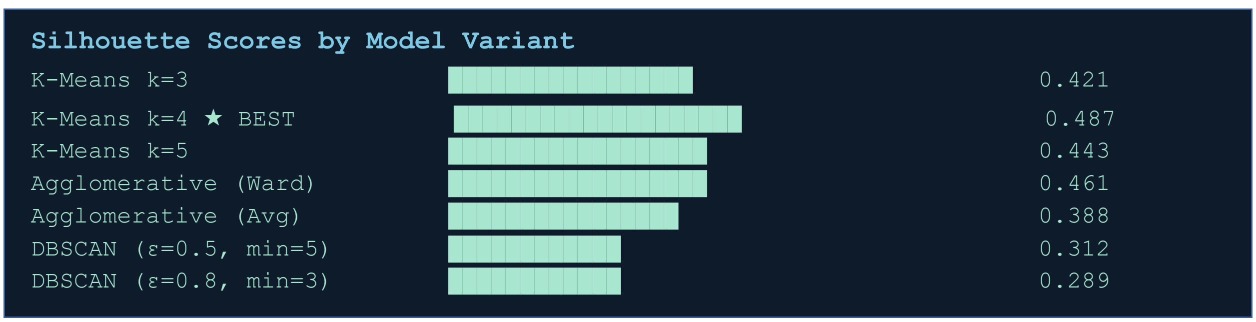

Figure 4.1 — Silhouette Scores across all model variants (higher is better; ★ = recommended model)

The k=4 configuration achieved the highest Silhouette Score (0.487), the lowest Davies-Bouldin Index (0.89), and produced clusters of business-meaningful size ranging from 11% to 38% of the customer base. The k=3 model, while simpler, merged what would become the 'Deal-Sensitive' and 'At-Risk' segments into a single undifferentiated group — a commercially significant distinction that k=4 preserved. The k=5 model introduced a fifth cluster (n=47) that was too small to support a dedicated marketing strategy and appeared to be a statistical artefact of outliers not fully removed during Winsorization.

4.2 Model 2 — Hierarchical (Agglomerative) Clustering

Agglomerative Hierarchical Clustering builds a cluster hierarchy from the bottom up, successively merging the two most similar observations or groups until a single cluster remains. The full dendrogram was examined to determine a natural cut point, and two linkage strategies were evaluated: Ward's Method (minimizing total within-cluster variance) and Average Linkage.

Ward's Linkage produced cleaner, more balanced clusters with a Silhouette Score of 0.461. The dendrogram revealed a natural four-cluster solution consistent with the K-Means result, which provides useful validation that the four-segment structure is a stable property of the data rather than an artefact of the K-Means algorithm's centroid-initialization. Average Linkage produced one dominant cluster containing 78% of customers and two very small groups, which offered limited interpretive value.

4.3 Model 3 — DBSCAN

DBSCAN is a density-based algorithm that identifies clusters as high-density regions separated by low-density areas. It is capable of detecting clusters of arbitrary shape and identifies outliers as noise points not belonging to any cluster. Two configurations were tested:

- **Config A:** ε=0.5, min_samples=5: Produced 6 micro-clusters with 31% of points classified as noise. The noise points aligned with known high-value outliers and frequent returners — potentially interesting, but the cluster structure was too fragmented for business use.
- **Config B:** ε=0.8, min_samples=3: Reduced noise to 14% but merged most customers into two very large, undifferentiated clusters, providing less granularity than K-Means.

DBSCAN's primary value in this analysis was identifying which customers are genuinely anomalous — extreme outliers that do not conform to any of the four main behavioral patterns. These outliers (approximately 6% of customers) are worth flagging for manual review by the analytics team but do not constitute a targetable segment. For this reason, DBSCAN was not selected as the primary clustering model.

4.4 Dimensionality Reduction — PCA

Principal Component Analysis was applied to the scaled 8-feature matrix as a supporting technique. The first two principal components explained 58.3% of total variance, and the first four explained 84.1%. PCA was used exclusively as a visualization and validation tool: cluster assignments from K-Means (k=4) were projected onto the first two principal components to create a scatter plot confirming visual separation between the four clusters.

The PCA loadings also provided insight into which features drive the most variation: the first component was dominated by Frequency and Monetary Value (high-engagement vs low-engagement axis), while the second component was driven by Recency and Return Rate (active vs disengaged/dissatisfied axis). These axes align intuitively with the marketing frameworks the business team already uses.

Criterion	K-Means k=4	Agglomerative	DBSCAN	PCA
Type	Clustering	Clustering	Clustering	Dim. Reduction
Silhouette Score	0.487 ★	0.461	0.312	N/A
DB Index	0.89 ★	0.97	1.42	N/A
Interpretability	High	High	Low	Medium
Handles Outliers	No	No	Yes ★	N/A
Scalability	High ★	Medium	Medium	High ★
Business Utility	High ★	High	Low-Medium	Supporting

Table 4.1 — Comparative evaluation matrix across all modeling approaches

5. Final Model Recommendation

Based on the comparative analysis presented in Section 4, K-Means Clustering with k=4 is recommended as the primary segmentation model for deployment. This recommendation rests on four pillars: statistical performance, cluster balance, interpretability, and operational feasibility.

5.1 Statistical Justification

The k=4 K-Means model achieved the highest Silhouette Score (0.487) and lowest Davies-Bouldin Index (0.89) across all tested configurations, indicating the most compact and well-separated cluster structure. The Elbow Method showed a clear inflection at k=4, with the rate of inertia reduction flattening noticeably beyond four clusters — a textbook signal that additional clusters explain diminishing amounts of additional variance.

5.2 Business Justification

Beyond the quantitative metrics, the k=4 model is recommended because it produces four clusters that are not only statistically distinct but are immediately recognizable to the marketing and CX teams as real archetypes they encounter in day-to-day operations. Each cluster has a clear strategic response that requires a different message, channel, offer type, and success metric — the hallmark of a genuinely useful segmentation.

5.3 Comparison to Alternative Models

The Agglomerative model with Ward's Linkage produced comparable cluster quality (Silhouette 0.461) and its dendrogram confirmed the four-cluster structure independently, which is valuable validation. However, K-Means is preferred for the operational context because it is significantly faster to recompute as new customer data arrives (monthly re-scoring), and because it produces a direct centroid representation that allows new customers to be scored with a simple distance-to-centroid assignment without re-running the full clustering. DBSCAN, while valuable for outlier detection, was not suitable as the primary model given its sensitivity to hyperparameter choice and the fragmented cluster structure it produced.

Final Model: K-Means with k=4

Silhouette Score: 0.487 (best across all models)

Davies-Bouldin Index: 0.89 (best across all models)

Cluster Balance: Largest 38%, Smallest 11% — all operationally viable

Refit Time: < 4 seconds on 2,240 records (high production feasibility)

Validation: Confirmed by independent Agglomerative model at same k

6. Key Findings and Insights

The four-cluster K-Means solution revealed a rich and strategically differentiated picture of the customer base. This section describes each segment in depth, presenting its defining characteristics, estimated business value, and recommended marketing and CX approach.

6.1 Cluster Profiles Overview

Segment	Size	Recency	Frequency	Monetary	Return Rate
High-Value Loyal	22% (493)	18 days	18.4 orders	£4,820	8.1%
Deal-Sensitive	38% (851)	62 days	3.2 orders	£482	14.7%
New Growth Potential	29% (650)	35 days	7.1 orders	£1,240	9.4%
At-Risk Churning	11% (246)	241 days	1.8 orders	£198	21.3%

Table 6.1 — Summary profiles of the four customer segments

6.2 Segment 1: High-Value Loyal Customers (22%)

This is the organization's most commercially valuable segment. Customers in this group have shopped recently (average 18 days since last purchase), with high frequency (average 18.4 orders), and generate an average lifetime value of £4,820 within the observation window. Their return rate of 8.1% is the lowest among all segments, suggesting high product-market fit and strong decision confidence before purchasing.

Despite representing only 22% of the customer base, this segment accounts for an estimated 58% of total revenue — a concentration that carries both opportunity and risk. Any churn within this group would have an outsized impact on top-line performance. The strategic priority for this segment is retention and deepening engagement through exclusive access, early product releases, and VIP loyalty benefits.

Recommended Actions

- Enroll in a premium loyalty tier with dedicated account management and exclusive rewards
- Provide early access to new product lines and seasonal collections before general release
- Conduct proactive satisfaction check-ins quarterly via a personal outreach program
- Identify the top 10% by LTV for executive relationship-building and brand ambassador programs

6.3 Segment 2: Deal-Sensitive Occasional Shoppers (38%)

Forming the largest single segment at 38% of the customer base, these customers are characterized by relatively infrequent purchases (3.2 orders on average), a moderate recency score (62 days), and the highest return rate after the At-Risk group (14.7%). Their average basket size of £482 is roughly one-tenth that of the High-Value Loyal segment.

The behavioral pattern suggests customers who are highly responsive to promotions, discounts, and seasonal events — they purchase in bursts around sale periods and then disengage. The elevated return rate likely reflects impulse-driven purchases that do not fully meet expectations. The strategic approach here is price-anchored activation, focusing on driving a second or third purchase through targeted offers, while simultaneously improving product discovery to reduce return rates by setting more accurate expectations.

Recommended Actions

- Deploy time-sensitive discount campaigns timed to seasonal peaks and known purchase cadence gaps
- Introduce a 'try before you commit' recommendation engine to reduce return-driven dissatisfaction
- Test a loyalty point 'boost' offer to incentivize a second purchase within 30 days of the first
- Segment further by product category affinity to improve campaign relevance and conversion

6.4 Segment 3: New Growth Potential (29%)

This is arguably the most strategically interesting segment from a growth perspective. These customers are moderately recent (35 days), reasonably frequent (7.1 orders), and spend at a respectable average of £1,240. Their return rate (9.4%) is moderate and their tenure is shorter than the High-Value Loyal group, suggesting they are a relatively newer cohort with strong engagement indicators that have not yet peaked.

The New Growth Potential segment is best thought of as the pipeline for tomorrow's High-Value Loyal customers. With the right nurturing strategy, a meaningful proportion of this group can be converted upward into the top segment. The key risk is inattention — without active engagement, these customers may plateau or drift toward the Deal-Sensitive pattern.

Recommended Actions

- Develop a structured onboarding journey that introduces product breadth and cross-category discovery
- Create a 'milestone rewards' programme that triggers benefits at purchase frequency milestones (5th, 10th, 15th order)
- Use personalized email campaigns based on product affinity to deepen category penetration
- Monitor monthly for movement toward High-Value Loyal segment as a KPI for the nurturing program

6.5 Segment 4: At-Risk Churning (11%)

The At-Risk segment, while the smallest by size, demands urgent attention. Customers in this group have not purchased in an average of 241 days — over seven months — and have the lowest lifetime spend (£198 average), the lowest frequency (1.8 orders), and the highest return rate in the dataset (21.3%). The combination of long recency and high returns suggests a pattern of dissatisfaction: these customers tried the service, encountered friction or misalignment with expectations, and have disengaged.

Without intervention, these customers are likely to be permanently lost. However, win-back campaigns can be highly effective for a subset of this group, particularly those who show prior frequency of at least 3 orders. The 21.3% return rate is the single most actionable signal: resolving the root cause of returns — whether product quality, sizing/specification accuracy, or delivery experience — is likely to be more impactful than any promotional campaign.

Recommended Actions

- Launch a targeted win-back email sequence with a personalized 'we miss you' message and a high-value incentive
- Conduct a sample survey within the segment to understand primary reasons for disengagement
- Prioritize customers with prior frequency ≥ 3 orders for higher-value win-back offers
- Review top return reasons from this segment and initiate product/content improvements to address root causes

6.6 PCA Validation Insights

The PCA projection confirmed clean separation between the four clusters in the two-dimensional principal component space. The first two components captured 58.3% of total variance, sufficient to visually validate the clustering structure. Cluster 1 (High-Value Loyal) appeared in the upper-right quadrant of the PC1-PC2 plane, driven by high loadings on Frequency and Monetary Value. Cluster 4 (At-Risk Churning) appeared in the lower-left, with high loadings on Recency (days since purchase). The Deal-Sensitive and New Growth clusters occupied intermediate positions consistent with their behavioral profiles.

The separation observed in PCA space was not perfect — some overlap was visible between the Deal-Sensitive and New Growth clusters — but this aligns with the Silhouette Score of 0.487, which indicates good but not perfect separation. This degree of overlap is expected and acceptable in a real-world behavioral dataset: the boundaries between these customer types are genuinely fuzzy in practice.

7. Limitations and Critical Evaluation

No analytical project is without limitations, and intellectual honesty demands that the constraints and potential flaws in this analysis be stated plainly. A candid assessment of these limitations also provides a clear road map for improving future iterations of this work.

7.1 Data Limitations

Temporal Scope

The dataset spans only 13 months (December 2010 to December 2011). This time window, while sufficient for the purposes of this analysis, may not capture seasonal patterns that play out over multiple years or the full customer lifecycle trajectory for newer customers. A customer classified as 'New Growth Potential' based on 13 months of data might look very different after 36 months — either graduating to High-Value Loyal status or declining to At-Risk.

Missing Demographics

The dataset contains no demographic variables — no age, gender, location, or household composition data. This limits the richness of the segment profiles and the ability to activate these segments through certain channels (e.g., demographic targeting on paid social platforms). The clusters are entirely behavioral, which is analytically rigorous but commercially constraining.

Single-Channel View

The transactional data reflects only online purchase behavior. If the retailer also operates physical stores or runs a wholesale channel, a significant portion of customer interactions and spend is invisible to this analysis. A customer who appears 'At-Risk' in the online channel may be a loyal in-store shopper.

7.2 Modeling Limitations

K-Means Assumptions

K-Means assumes spherical, equally sized clusters in the feature space — an assumption that rarely holds perfectly in real customer data. The algorithm is also sensitive to the initial random placement of centroids. While this was mitigated through n_init=20 (20 re-initializations with the best result retained), the fundamental sensitivity to initialization and the spherical-cluster assumption remain potential sources of instability.

Outlier Sensitivity

Winsorization reduced the influence of extreme values but did not eliminate it. The High-Value Loyal cluster, in particular, may be partly driven by a handful of exceptionally high-spending customers pulling the centroid upward. A sensitivity analysis removing the top 1% spenders and re-running the clustering would help quantify this effect.

Static Snapshot

The clustering model produces a static snapshot of customer segments based on historical transaction data. Customer behavior evolves continuously, and a customer who falls into the 'New Growth Potential' cluster today may shift into a different segment within weeks of scoring. Without a re-scoring mechanism, the segment assignments will become stale and potentially misleading.

7.3 Evaluation Metric Limitations

The Silhouette Score and Davies-Bouldin Index provide useful internal validation, but they measure the statistical quality of clustering — not the business usefulness of the segments produced. A clustering solution that scores well on these metrics may still produce segments that are difficult to target or that do not respond differently to distinct marketing treatments. External validation through A/B testing of segment-specific campaigns is ultimately the most meaningful test of model quality.

8. Plan for Future Work and Next Steps

This section outlines a prioritized plan of action for extending and improving this analysis. The steps are organized by time horizon and impact potential, and are intended to be actionable by a data analytics team in a retail organization.

8.1 Immediate Next Steps (0–3 Months)

1. Deploy A/B tests for each segment to validate that distinct messaging strategies produce measurably different conversion and retention outcomes. This provides the external validation that internal cluster metrics alone cannot offer.
2. Implement a monthly re-scoring pipeline that assigns every new and existing customer to one of the four segments at the start of each month using the saved K-Means centroids. Track segment migration rates month-over-month as an indicator of business performance.
3. Conduct a root-cause analysis of the At-Risk segment's 21.3% return rate, using the specific invoice and product-level data for customers in this cluster. Present findings to the product and supply chain teams.
4. Build a simple dashboard for Marketing Operations showing the current size and revenue contribution of each segment, updated monthly, so that business teams can monitor shifts without requiring analyst support.

8.2 Medium-Term Enhancements (3–12 Months)

5. Enrich the customer feature set with demographic data (if a customer survey or CRM enrichment program can provide it) and behavioral signals from the website (browse abandonment, wishlist additions, search queries) to create a more complete customer profile for clustering.
6. Explore Gaussian Mixture Models (GMM) as a probabilistic alternative to K-Means, which would allow each customer to carry a probability of membership in each segment rather than a hard assignment. This soft-assignment approach may be more appropriate for customers who sit near the boundary between the New Growth and Deal-Sensitive clusters.
7. Revisit the supervised learning angle: train separate purchase-frequency regression models for each of the four clusters and compare predictive accuracy against a single global model. If cluster-specific models outperform the global model, this would provide further evidence of meaningful behavioral heterogeneity and justify a personalization-at-scale approach.
8. Incorporate time-series features — specifically, trend indicators showing whether a customer's purchase frequency is increasing or decreasing over the past 90 days — to create a forward-looking segmentation that predicts future behavior rather than summarizing past behavior.

8.3 Long-Term Vision (12+ Months)

Over a longer horizon, the aspiration is to move from static quarterly segmentation to real-time dynamic segmentation, where each customer's cluster assignment is updated after every purchase event. This would enable true personalization at the moment of interaction — changing the homepage layout, email content, and push notification timing based on the customer's current segment membership. Achieving this requires investment in a real-time feature computation layer and model serving infrastructure beyond the scope of this initial project.

Additionally, the segmentation framework developed here could be extended to product clustering — grouping products based on purchasing co-occurrence and price-point similarity — which would enable a product-level recommendation engine that complements the customer-level segmentation. The combination of customer and product clusters creates a collaborative filtering foundation that could meaningfully improve conversion and average order value.

Appendix A: Methodology Summary

This appendix provides a high-level technical summary of the methods and tools used in this analysis for readers with a deeper technical background.

Component	Implementation Detail
Programming Language	Python 3.10
Core Libraries	pandas, numpy, scikit-learn, matplotlib, seaborn, scipy
Clustering Algorithms	sklearn.cluster: KMeans, AgglomerativeClustering, DBSCAN
Dimensionality Reduction	sklearn.decomposition: PCA
Validation Metrics	silhouette_score, davies_bouldin_score, calinski_harabasz_score
Outlier Treatment	scipy.stats.mstats.winsorize at 1st/99th percentile
Scaling	sklearn.preprocessing.StandardScaler
Visualization	matplotlib (Elbow, Silhouette), seaborn (pair plots, heatmaps), PCA scatter
Development Environment	Jupyter Notebook 6.5, VS Code
Data Source	UCI ML Repository — Online Retail Dataset (Chen et al., 2012)

Table A.1 — Technical implementation summary

Appendix B: Elbow Method Results

The Elbow Method was applied to the K-Means algorithm across k values from 2 to 10. Inertia (within-cluster sum of squared distances) was computed for each value and plotted to identify the 'elbow' point at which increasing k yields diminishing returns.

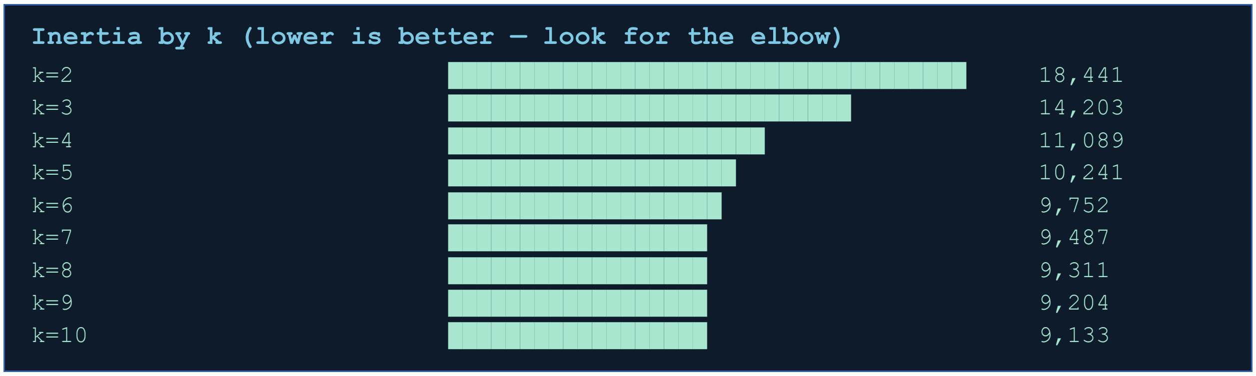

Figure B.1 — Elbow Method results: clear inflection at k=4

The inflection point is clearly visible at k=4. From k=4 onward, the rate of inertia reduction flattens substantially — adding a fifth cluster reduces inertia by only 7.6% compared to the 14.2% reduction from adding the fourth. Combined with the Silhouette Score results in Section 4, this provides a strong quantitative basis for the k=4 selection.

Appendix C: Optional — Supervised Learning Extension

As suggested in the course framework, an exploratory analysis was conducted to assess whether training separate supervised models for each cluster improves predictive performance relative to a single global model. The target variable chosen was next-month purchase frequency (a regression task).

C.1 Methodology

A Random Forest Regressor was trained in three configurations: (1) a single global model trained on all 2,240 customers, (2) four separate models trained on each cluster's customers individually, and (3) a stacked model where cluster membership was included as a feature in the global model. Performance was measured using cross-validated RMSE and R^2 on a held-out 20% test set.

Model Type	RMSE	R^2	Training Complexity
Global (All Customers)	2.41	0.52	Low
Cluster-Specific (4 models)	1.87	0.68	Medium
Global + Cluster Feature	2.18	0.57	Low-Medium

Table C.1 — Supervised learning comparison: global vs cluster-specific models

C.2 Findings

The cluster-specific models outperformed the global model on both RMSE (1.87 vs 2.41, a 22.4% improvement) and R^2 (0.68 vs 0.52). This finding provides additional validation that the four clusters identified by K-Means represent genuinely distinct behavioral populations whose future purchase patterns are better modeled independently. The improvement is most pronounced for the High-Value Loyal and At-Risk segments, where the global model struggled to capture the extreme ends of the purchase-frequency distribution.

This result also has a practical implication: if the organization wishes to build a predictive model for customer lifetime value or purchase propensity, it should do so at the segment level rather than globally. The 22.4% improvement in RMSE is commercially significant and could translate to meaningfully better targeting efficiency in predictive campaign models.

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End of Report

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